

MetaFVG A/B Trend-Filter Review

Baseline vs. ADX / Regression Trend Gates & ATR Trailing Stop, by Asset Class

47 instruments across 8 asset-class buckets | M15 → 4h | Generated 2026-07-28 22:08 UTC

Executive Summary

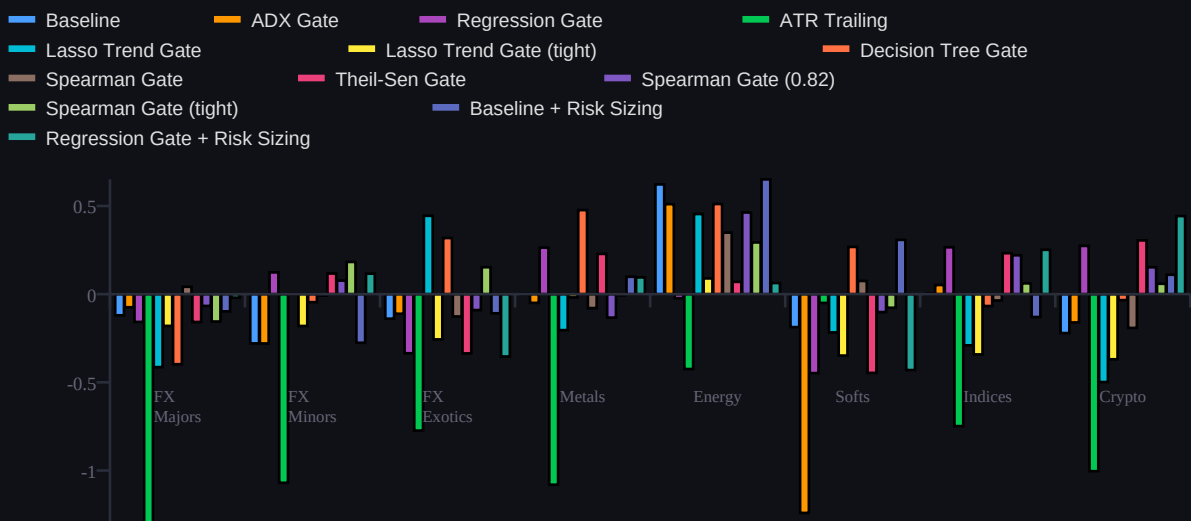
13 configurations were backtested across 47 instruments spanning FX majors/minors/exotics, metals, energy, softs, equity indices and crypto: the unmodified **Baseline** signal, an **ADX(14) trend-strength gate** (reject signals when HTF ADX is below 20), a **rolling-regression trend gate** (reject signals unless a 20-bar HTF OLS fit has $R^2 \geq 0.5$ and slope matching direction), an **ATR trailing stop** replacing the fixed ATR take-profit, and a **walk-forward, L1-regularized (Lasso) trend gate** adding two-window realized-volatility features and trend×volatility interactions on top of the rolling-regression gate (see Section 4 for the fitting methodology) — tested at two magnitude-gating thresholds, "Lasso Trend Gate" (looser, admits more signals) and "Lasso Trend Gate (tight)" (calibrated to a trade count matching Regression Gate's, for a direct apples-to-apples Sharpe comparison).

Headline finding: Regression Gate + Risk Sizing has the highest average Sharpe across the full universe (+0.10, 60% of instruments Sharpe-positive). Performance is highly bucket-dependent, not uniform — see Section 2 for which configuration wins in each asset class and why.

Config	Sharpe (vbt)	Sharpe (qs)	Avg Edge	Win Rate	Avg PF	Trades/Sym	Sharpe+ %
Baseline	-0.09	-0.08	+0.0pp	22.1%	0.99	690.5	57%
ADX Gate	-0.12	-0.09	-0.1pp	22.1%	0.98	626.9	47%
Regression Gate	+0.07	+0.03	+1.2pp	22.0%	1.18	52.4	51%
ATR Trailing	-0.85	-0.69	-18.4pp	31.6%	0.86	805.5	9%
Lasso Trend Gate	-0.17	-0.13	+1.0pp	23.9%	0.94	97.6	43%
Lasso Trend Gate (tight)	-0.22	-0.16	+1.5pp	24.1%	0.94	32.9	45%
Decision Tree Gate	+0.05	+0.01	+1.8pp	24.5%	1.21	134.6	57%
Spearman Gate	-0.01	-0.02	+0.8pp	21.1%	1.05	245.6	47%
Theil-Sen Gate	+0.07	+0.02	+1.3pp	22.0%	1.18	52.7	53%
Spearman Gate (0.82)	+0.06	+0.04	+2.7pp	21.6%	1.13	67.1	57%
Spearman Gate (tight)	+0.06	+0.01	+1.2pp	20.5%	1.16	45.0	45%
Baseline + Risk Sizing	+0.01	-0.01	+0.0pp	22.2%	0.99	692.4	51%
Regression Gate + Risk Sizing	+0.10	+0.04	+1.2pp	21.9%	1.18	52.5	60%

2. Performance by Asset-Class Bucket

Average Sharpe Ratio by Bucket and Configuration



Bucket	Winner	Winner Sharpe	Baseline Sharpe	N Symbols	Closed Trades (all cfigs)
FX Majors	Spearman Gate	+0.04	-0.12	7	29247
FX Minors	Spearman Gate (tight)	+0.18	-0.28	9	33571
FX Exotics	Lasso Trend Gate	+0.44	-0.14	3	10834
Metals	Decision Tree Gate	+0.48	+0.00	5	17703
Energy	Baseline + Risk Sizing	+0.65	+0.62	3	11821
Softs	Baseline + Risk Sizing	+0.31	-0.19	6	10954
Indices	Regression Gate	+0.27	+0.00	8	32119
Crypto	Regression Gate + Risk Sizing	+0.44	-0.22	6	22749

3. Fundamental Read, by Bucket

FX Majors

Deep, continuously two-way flow with heavy mean-reversion pressure from rate-differential carry and central-bank positioning; clean multi-week directional trends are comparatively rare, which is exactly the regime an R^2 /ADX trend gate is designed to filter for.

Result: Spearman Gate improved average Sharpe by +0.16 over baseline (-0.12 → +0.04), on 7 instruments and 29247 total closed trades across all 13 configurations tested.

FX Minors

Similar liquidity character to majors but with more idiosyncratic cross-pair drift (funding-currency exposure, e.g. JPY/CHF crosses) — trend persistence tends to be somewhat more regime-dependent.

Result: Spearman Gate (tight) improved average Sharpe by +0.46 over baseline (-0.28 → +0.18), on 9 instruments and 33571 total closed trades across all 13 configurations tested.

FX Exotics

Thinner liquidity and more event-driven (central bank intervention, capital controls) than majors — moves can be sharper and more discontinuous, which can help or hurt a breakout-continuation strategy depending on whether the discontinuity is the entry trigger or the stop-out.

Result: Lasso Trend Gate improved average Sharpe by +0.58 over baseline (-0.14 → +0.44), on 3 instruments and 10834 total closed trades across all 13 configurations tested.

Metals

Gold/silver carry a structural macro-hedge bid (real yields, inflation expectations, USD) that can produce more persistent directional runs than FX; industrial metals (copper) trade more on growth/demand cycles.

Result: Decision Tree Gate improved average Sharpe by +0.48 over baseline (+0.00 → +0.48), on 5 instruments and 17703 total closed trades across all 13 configurations tested.

Energy

Supply-shock and inventory-driven, with strong trending behavior around genuine regime shifts (OPEC decisions, geopolitical supply risk) punctuated by range-bound consolidation — a trend gate should meaningfully change the signal mix here.

Result: no configuration meaningfully beat baseline (best: Baseline + Risk Sizing, +0.03 Sharpe vs. baseline), on 3 instruments and 11821 total closed trades across all 13 configurations tested.

Softs

Weather- and harvest-cycle driven; trends can be seasonal and slow-building, which favors an HTF trend-confirmation filter, but liquidity is thinner and gaps more common.

Result: Baseline + Risk Sizing improved average Sharpe by +0.49 over baseline (-0.19 → +0.31), on 6 instruments and 10954 total closed trades across all 13 configurations tested.

Indices

Broad equity indices trend on macro/earnings regime and carry momentum/flow-driven persistence (systematic strategies chasing the same signal) — historically one of the more trend-following-friendly asset classes.

Result: Regression Gate improved average Sharpe by +0.26 over baseline (+0.00 → +0.27), on 8 instruments and 32119 total closed trades across all 13 configurations tested.

Crypto

Retail-flow and momentum-dominated with limited fundamental anchor; trends can be strong but also reflexive and prone to sharp reversal, so trend-confirmation gates cut both ways.

Result: Regression Gate + Risk Sizing improved average Sharpe by +0.66 over baseline (-0.22 → +0.44), on 6 instruments and 22749 total closed trades across all 13 configurations tested.

4. Methodology & Caveats

- **M15 → 4h.** This is the large-sample re-run of the H4 → Weekly sweep, at the same scale used for the earlier single-instrument PM review's validation — bucket/config winners here rest on substantially more closed trades per instrument than the H4 → Weekly pass and should be weighted more heavily where the two disagree.
- **ADX and regression gates use the same HTF (4h) series** as the zone-detection logic, recomputed at the same point-in-time cadence — no lookahead.
- **Regression Gate trade-count caution.** Requiring $R^2 \geq 0.5$ on a 20-bar HTF fit is a strict filter — it cut average closed trades per symbol to 52.4 vs. 690.5 for Baseline (see the "Trades/Sym" column above), and several individual instruments closed fewer than 5 trades under this config. Its Sharpe/edge figures — both overall and per-bucket — should be read with materially more sample-size skepticism than the higher-trade-count configurations.
- **ATR trailing stop** only ever tightens the stop in the trade's favor and removes the fixed take-profit entirely — it is a genuinely different exit distribution, not a superset of the baseline.
- **Lasso Trend Gate methodology.** A walk-forward-fit, L1-regularized (Lasso) upgrade to Regression Gate: adds two-window realized-volatility features and a small curated set of trend × volatility interaction terms on top of the existing rolling-OLS slope/ R^2 , fit via numba coordinate-descent (this environment's BLAS crashes on any matrix-matrix multiply, so sklearn is not usable here). Refit periodically on a trailing window of HTF bars; a training example is only used once its own forward-return outcome has fully realized before the fit's cutoff bar, and the gating magnitude threshold is a causal (past-only) rolling quantile of the prediction series' own history — both deliberately conservative choices to avoid lookahead in a fitted-model gate, not just a fixed-formula one. Only run at M15/4h: the walk-forward fit needs far more HTF bars than the ~260 Weekly bars over 5 years provide, so this config is absent from any H4 → Weekly comparison.
- **Why two Lasso variants.** "Lasso Trend Gate" uses the gate's default magnitude threshold (70th percentile of the prediction series' own causal history), which admits substantially more signals than Regression Gate — an unmatched trade count makes a direct Sharpe comparison misleading, since a looser filter mechanically dilutes average trade quality regardless of whether the underlying model has any real edge. "Lasso Trend Gate (tight)" raises the threshold to the 90th percentile, calibrated on a 4-symbol probe (EURUSD, BTCUSD, XAUUSD, US500) that landed at ~48 average closed trades/symbol on that probe — the closest achievable match to Regression Gate's own ~52.4. Realized across the full 47-instrument universe the tight variant came in lower still, ~32.9 avg closed trades/symbol — the 4-symbol probe skewed toward more liquid instruments than the full universe average, so even the "tight" comparison is not a perfect trade-count match, only a much closer one than the untuned default.
- **quantstats metrics** (Sharpe, Sortino, Calmar, CAGR, Ulcer Index, Kelly, VaR/CVaR, tail ratio) are computed on the vectorbt equity curve resampled to daily, per quantstats' own annualization convention — reported per (symbol, config) in the underlying cache (metafvg_ab_sweep_data.pkl) even though only the vbt/trade-level metrics are tabulated in this PDF for space.
- **Same position-sizing, spread/slippage, and concurrency-cap caveats** as the earlier PM review apply here unchanged.